JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI VENKATESHWARA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

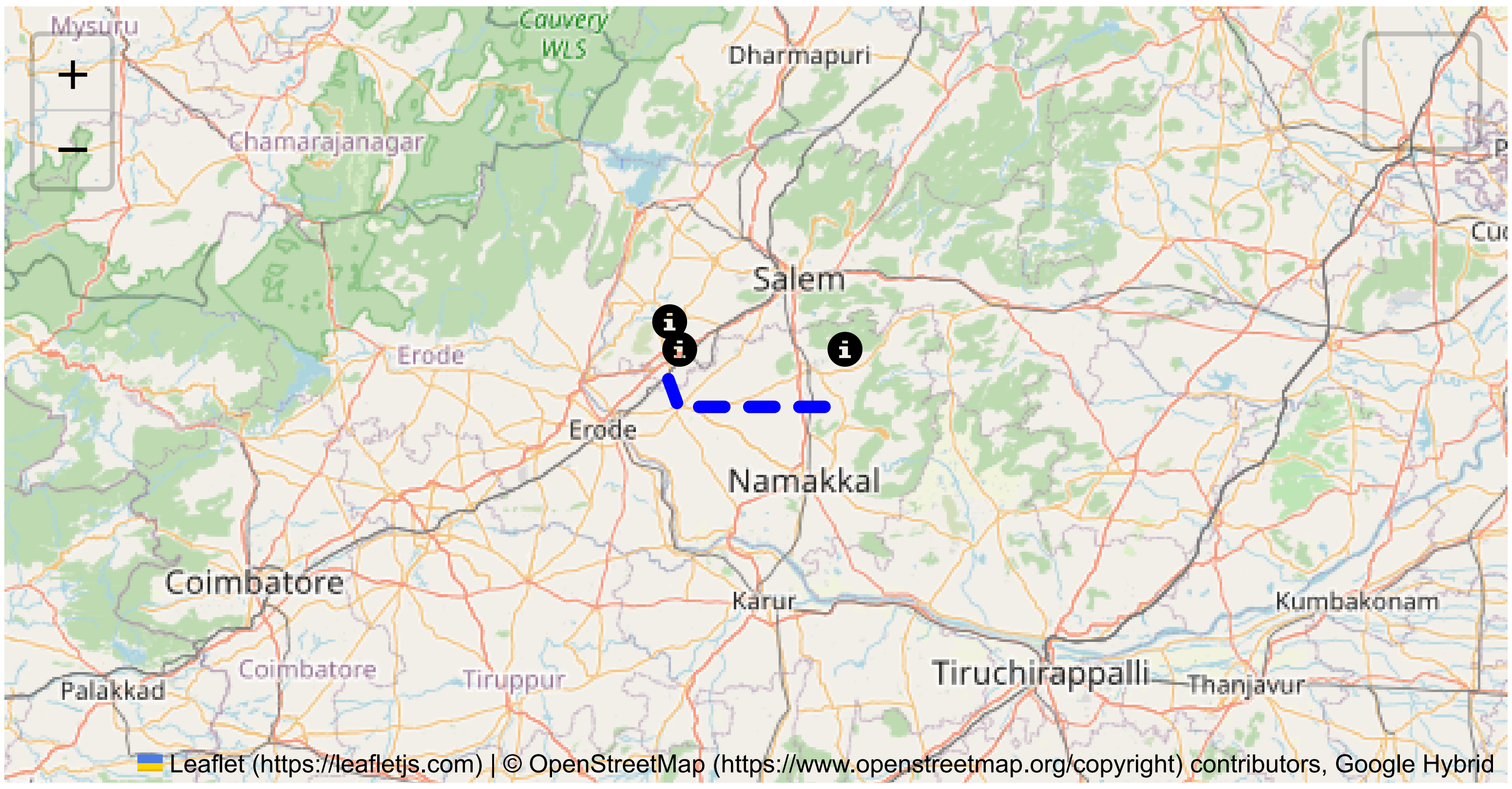
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

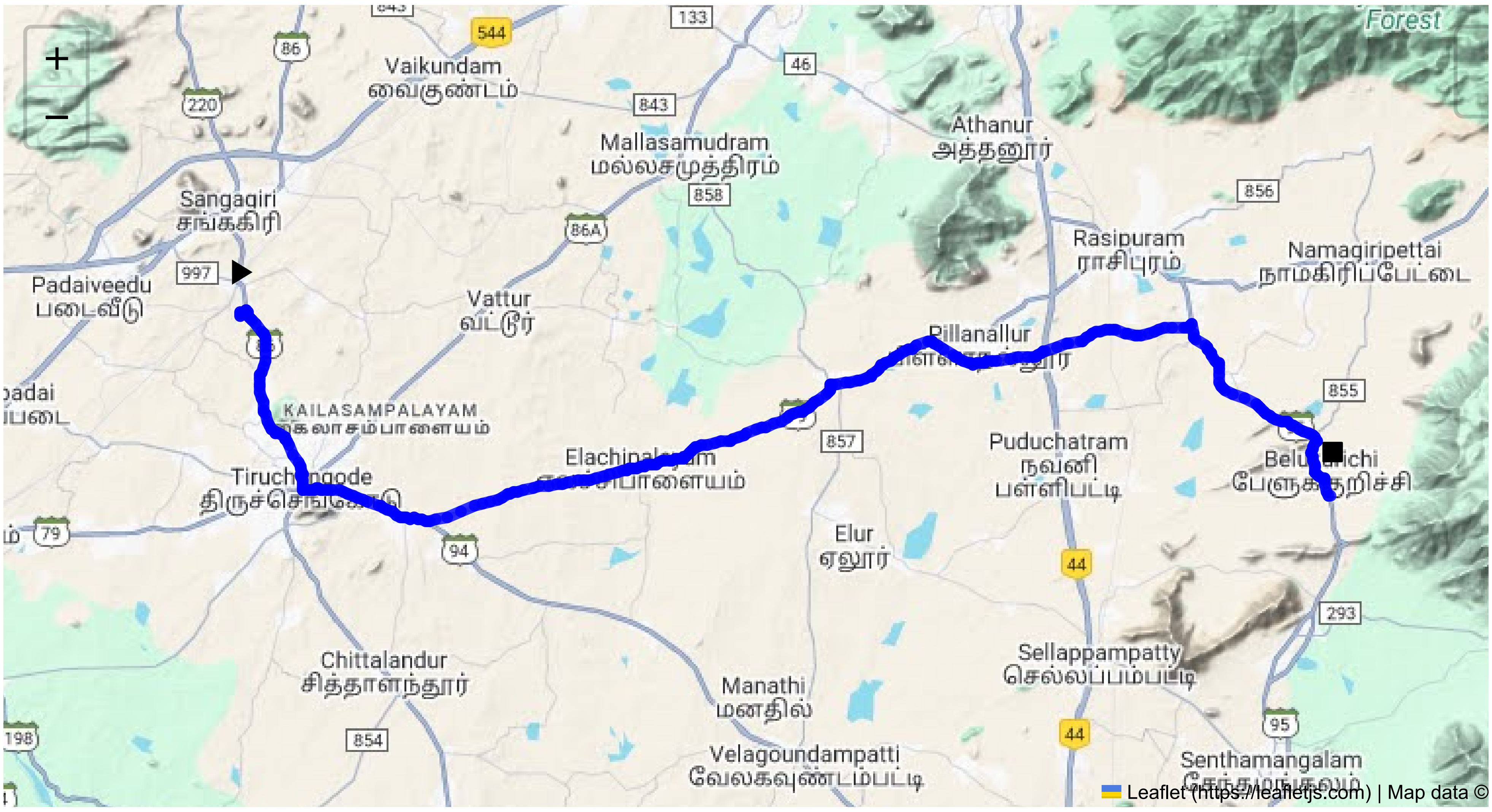
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 54.34 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.376412, 78.256172)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The specified route starts at CVQF+23W, Sangagiri, passes through S Car St, Kamalar, Tiruchengode, and ends at 97G4+F9G, SH 95. The journey spans approximately 54.34 kilometers. The roads primarily consist of state highways and local roads, intersecting smaller towns and rural areas.

2. Typical Weather Conditions and Hazards

Tamil Nadu generally has a tropical climate. The area experiences:

- **Summer (Mar to May):** High temperatures (up to 38°C), leading to potential heat-related vehicle issues.
- **Monsoon (Jun to Sep):** Heavy rainfall can cause slippery roads and localized flooding.
- **Winter (Oct to Feb):** Mild with occasional rain; fog might reduce visibility early morning. Weather-related hazards include possible waterlogging on highways during monsoon and reduced visibility due to fog.

3. Traffic Patterns

- **Peak Hours:** 8-10 AM and 5-7 PM, particularly around towns like Tiruchengode where local traffic is heavier.
- **Congestion-Prone Areas:** Major intersections in Tiruchengode may see slow-moving traffic. Markets and school zones could add unexpected delays.

4. Road Quality and Infrastructure

- **State Highways:** Generally well-maintained, but watch for wear-and-tear post-monsoon.
- **Local Roads:** Possible issues include potholes and pedestrian traffic in villages.
- **Signage and Lighting:** Generally adequate on highways, but limited lighting in rural stretches could pose issues at night.

5. Alternative Routes for Emergencies

- Diversion through NH44 can be considered if the primary route is inaccessible.
- Local roads to neighboring towns like Namakkal may offer alternative paths, but check the road conditions beforehand.

6. Local Regulations Affecting Hazardous Material Transport

- Hazardous materials are generally restricted during peak hours in populated areas.
- Specific permits may be required for certain materials; consult local transport authorities for detailed regulations.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Limited records of major incidents, but caution advised during monsoon seasons due to increased accident risk from slippery roads.
- Local reports suggest periodic truck-related mishaps near market areas due to congestion.

8. Environmental Considerations and Sensitive Areas

- Proximity to agricultural fields; spillage could impact crop health.
- Passes through or nearby to water bodies—ensure vehicles are in good condition to prevent pollution.

9. Communication Coverage

- **General Coverage:** Good coverage in towns and along main highways.
- **Potential Dead Zones:** Rural stretches may have weak signal and potential lost connectivity.

10. Estimated Emergency Response Times

- **Urban Areas (Tiruchengode):** 20-30 minutes for emergency services.
- **Rural Areas:** Up to an hour depending on road conditions and weather.

12. Overall Summary of Risk Assessment

The route is generally manageable, with standard precautions necessary for heavy vehicle operations. The primary concerns stem from weather-related issues and rural infrastructure quality. Actionable risk mitigation strategies include planning around peak hours, having emergency contact information readily available, and ensuring thorough vehicle checks before departure. Communication gaps and environmental effects warrant particular attention for improving overall safety. Efficient transport of hazardous materials requires adherence to local regulations and enhanced vigilance given the presence of sensitive zones.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	Medium	11.37809, 77.89834	30 KM/Hr
6	Turn	High	11.41466, 78.08070	15 KM/Hr
7	Turn	High	11.42962, 78.11578	15 KM/Hr
8	Turn	Medium	11.42450, 78.15278	30 KM/Hr
9	Blind Spot	Blind Spot	11.43542, 78.20754	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.43035, 78.20865	30 KM/Hr
11	Turn	Medium	11.42167, 78.21705	30 KM/Hr
12	Turn	Medium	11.39642, 78.25111	30 KM/Hr
13	Turn	Medium	11.39536, 78.25156	30 KM/Hr
14	Turn	Medium	11.38226, 78.25407	30 KM/Hr
15	Turn	High	11.37644, 78.25558	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
9	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
10	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
12	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37809, 77.89834

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41466, 78.08070

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42962, 78.11578

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42450, 78.15278

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.43542, 78.20754

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43035, 78.20865

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42167, 78.21705

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39642, 78.25111

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39536, 78.25156

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38226, 78.25407

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37644, 78.25558